Applied Energy 372 (2024) 123770

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

Enhancing PV hosting capacity and mitigating congestion in distribution networks with deep learning based PV forecasting and battery management

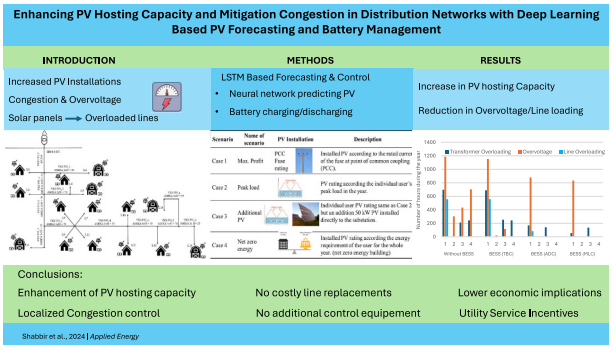
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HIGHLIGHTS

- Identified overloaded lines and buses using a bus importance index solution.
- Evaluated three congestion control schemes: BTC, ADC, and MLC.
- Conducted detailed analysis on transformer and line overloading.
- Concluded ML-based BESS control as the most economical and effective solution.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Renewable energy
Congestion control
Hosting capacity
Deep learning
LSTM

ABSTRACT

The extensive deployment of domestic photovoltaic (PV) systems may result in exceeding the limits of the network's PV hosting capacity (HC), which leads to energy delivery congestion and overvoltage problems in low voltage (LV) networks. These problems amplify at times when PV energy generation is near its peak and the domestic load is at its lowest. The convenient solution is to upgrade powerlines or utilize external mitigation devices, both are costly and reconstructing all the networks is a challenging task. Therefore, in this study, a machine learning-based control (MLC) solution is proposed to relieve the overloading of the lines and overvoltage problems caused by the high penetration of PV installations without costly line replacements or additional devices. This MLC technique employs a long short-term memory (LSTM) algorithm for a day ahead PV forecast incorporated to decide the charging/discharging of the residential battery energy storage systems (BESS). The study is conducted based on a real-life LV distribution system with 15 households in the network segment by considering the measured data of actual residential loads and installed PV energy generation for the entire year. Four different rated PV system installation scenarios are investigated through power flow analysis. Moreover, three different BESS utilization techniques are compared along with traditional peak power curtailment (PPC) and reactive power control (RPC) techniques. The MLC approach shows a 30% and 4% lower number

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https://doi.org/10.1016/j.apenergy.2024.123770
Received 25 March 2024; Received in revised form 27 May 2024; Accepted 18 June 2024
Available online 4 July 2024
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of overvoltage hours compared to analytics-driven control (ADC) and trivial battery control (BTC) techniques, respectively for excessive PV installations (case 1). The MLC methods address the HC in a more dedicated manner, eliminating congestion problems in PV installation according to the peak load (Case 2 & 3) and net zero energy (Case 4). The techno-economic analysis indicates that the MLC technique with local grid-supporting capabilities increases the HC and reduces overloading and overvoltage problems in the LV network in all cases. The customers receive 8% to 22% more in savings as compared to TBC control technique and could also get further lower energy costs from the utilities for grid service.

Nomenclature

Abbreviations

ADC	Analytics Driven Control
BESS	Battery energy storage systems
CoE	Cost of Energy
DL	Deep learning
DoD	Depth of discharge
DN	Distribution network
DR	Demand Response
DSO	Distribution system operator
EMS	Energy management system
ER	Energy router
EU	European union
HC	Hosting capacity
LP	Linear programming
LSTM	Long short-term memory
LV	Low voltage
ML	Machine learning
MLC	Machine learning based control
MPC	Model predictive control
MV	Medium voltage
nZEBs	Nearly zero-energy buildings
OLTCs	On-load tap changer
QoS	Quality of service
PPO	Proximity policy optimization
PPC	Peak Power Curtailment
PV	Photovoltaic
RES	Renewable energy sources
RL	Reinforcement learning
RO	Robust optimization
RNN	Recurrent neural networks
RPC	Reactive power control
SO	Stochastic programming
SoC	State of charge
TBC	Trivial battery control
TSO	Transmission system operator

Symbols

$\boldsymbol{A_{ij}}$	Adjacency matrix
$\boldsymbol{C_{bat}^{ch,u}}$	Battery charging cost per unit
$\boldsymbol{C_{bat}^{dis,u}}$	Battery discharging cost per unit
$\boldsymbol{C_{grid}^{sell,u}}$	Per-unit energy selling cost
$\boldsymbol{C_{PV}^{u}}$	Per-unit energy selling cost
$\boldsymbol{C_{Bat}}$	Cost of battery
$\boldsymbol{C_{Inv./Unit}}$	Per-unit inverter cost
$\boldsymbol{C_{PV}}$	Cost of PV
$\boldsymbol{DoD}$	Depth of discharge
$\boldsymbol{E_{BC}}$	Battery capacity
$\boldsymbol{E_{Day}}$	Average daily energy consumption
$\boldsymbol{E_{bat}^{ch}}$	Energy received from the battery
$\boldsymbol{E_{bat}^{dis}}$	Energy supplied from the battery
$\boldsymbol{E_{grid}^{sell}}$	Energy sold to the grid
$\boldsymbol{E_{grid}^{pur}}$	Energy purchased from the grid
$\boldsymbol{E_L}$	Electrical energy consumption
$\boldsymbol{st}$	Time interval
$\boldsymbol{\eta}$	BESS efficiency
$\boldsymbol{i_t}$	Input gate
$\boldsymbol{o_t}$	Output gate
$\boldsymbol{C_{t-1}}$	Input from memory cell
$\boldsymbol{E_{bat}^{max}}$	Max. energy from the battery
$\boldsymbol{E_{PV}}$	PV generated energy
$\boldsymbol{E_{PV}^{max}}$	Max. energy from PV
$\boldsymbol{INV_{rated}}$	Rated inverter power
$\boldsymbol{K_p}$	Status of the BESS
$\boldsymbol{P_{rated}}$	Rated PV power
$\boldsymbol{P_i}$	Actual power
$\boldsymbol{P_{max}}$	Max. Power flow
$\boldsymbol{P_{ratio}}$	Power flow ratio
$\boldsymbol{C_j}$	Bus centrality
$\boldsymbol{k_j^{out}}$	Out Outdegree of the bus
$\boldsymbol{\hat{C}_t}$	Newly created vector
$\boldsymbol{D}$	Diagonal matrix
$\boldsymbol{P_z}$	Maximum charging power
$\boldsymbol{\sigma}$	Sigmoid function
$\boldsymbol{w}$	Respective gate neuron
$\boldsymbol{x_t}$	Input signal
$\boldsymbol{h_t}$	Output
b	Gate bias
t	Timestamp
$\boldsymbol{f_t}$	Forgetting state

1. Introduction

The installation of solar photovoltaic (PV) systems at the household level has gained momentum as a major clean and renewable energy resource (RES), providing affordable and sustainable energy [1,2]. However, the growing participation of PV systems in residential areas can lead to technical problems for the electric energy distribution networks (DN) [3]. For instance, in a residential district, the solar radiation remains the same for most customer points at the same time, which means that the PV-produced energy is fed to the DN at near peak power at the same time from different households [4]. Therefore, the excessive and unplanned installation of the PV system and its grid connectivity can turn its eco-friendly advantage into disadvantages with many possible operational issues in the DN [5], such as the risk of overvoltage, overloaded lines due to increased reactive power flow can impact the power system protection schemes [6,7].

In recent times, researchers have introduced aspects of the evaluation of the hosting capacity (HC) of the network due to solar PV installations at particular locations or distribution lines due to its essence [8]. Any DN can face operational limitations if the solar PV penetration

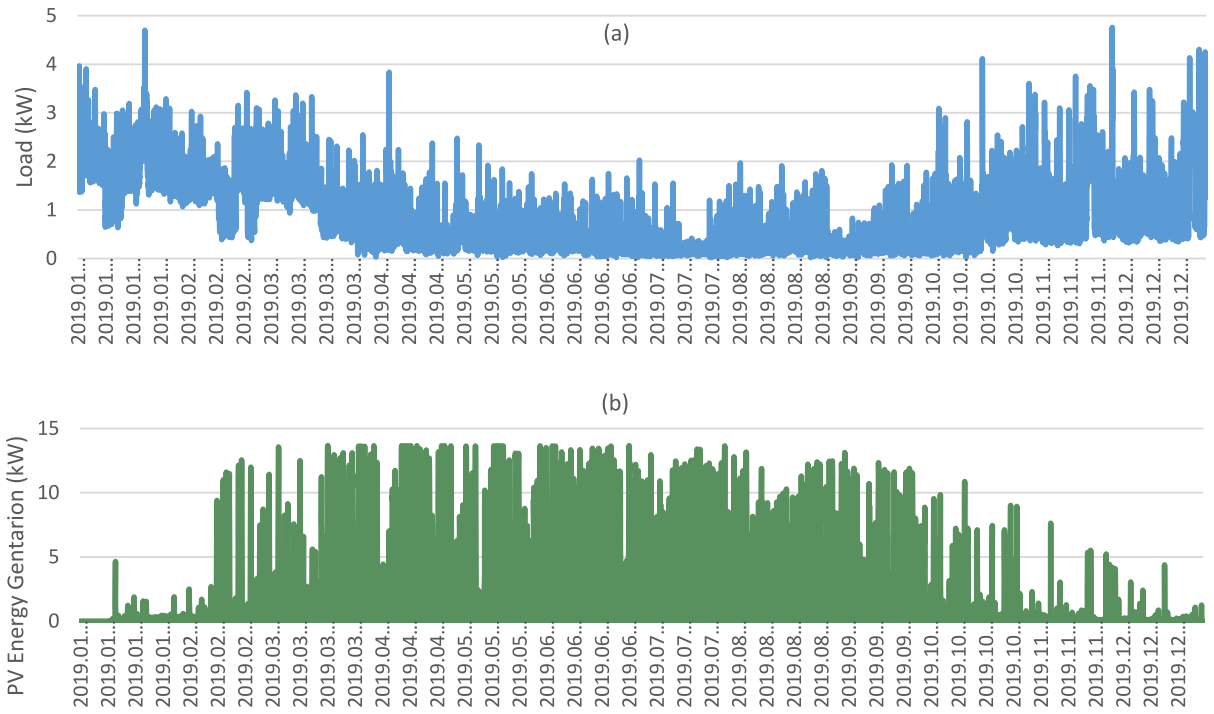


Fig. 1. (a) the residential load (b) PV energy generation throughout the year.

exceeds the rated infeed level, known as the HC of the network, and can jeopardize the safety and reliable operation of the grid [9]. For a distribution system operator (DSO), improving system HC is an essential goal while maintaining quality of service (QoS) by the IEC/EN/IEEE Standards [10]. Part of the HC depends on two major factors, such as the thermal-rated throughput capacity of the distribution feeder and the feeder's voltage profile. There are a variety of parametric definitions for defining HC, such as maximum photovoltaic energy generation, maximum feeder ratio load [6], production and consumption [11], and transformer rating [12]. In the real residential area network, the HC value also depends on solar PV energy generation and the self-consumption of the residential user [13]. Several research works have also been published at medium voltage (MV) levels [14–16].

In low-voltage (LV) DNs, the rooftop solar PV systems are gradually spreading, one reason being the nearly-zero energy buildings (nZEBs) concept where all energy utilized should be produced on-site. Commonly in such systems, PV installed on the rooftop is connected to the LV AC distribution network through a grid-tied inverter [17]. An overvoltage situation can quickly arise in the LV networks when the solar PV systems are operating at their maximum limit during peak generation hours (middle hours of the day) and local residential load/consumption is very low. The problem is depicted in Fig. 1 by showing the residential load and PV energy generation throughout the year of an Estonian household. In many areas of the world where peak loads, such as air conditioners are used during the daytime, the HC of the PV system can be a lighter problem. However, in other regions without summer daytime local load, the peak power injection causes overvoltage in the network due to low load, power reversal, and disturbance in the reactive power [18]. The operation of the LV grid within the standardized power quality limits, such as EN50160 or IEEE 519, becomes a problem as the distribution lines have a limited capacity to support the expected voltage drop/rise [19,20]. Another factor that limits HC is the load current capacity of the distribution lines [21]. Compared to voltage level violation which is a major concern in rural and suburban networks, line overload is a greater problem in urban networks [22,23]. To keep the operating margins, the network operator must resort to a very costly solution, replacing the line or line segments with a higher load-capable cable. Utility operators must upgrade overloaded components or use some smart solution to take advantage of problems of overvoltage on the electrical line and transformer and an increase in voltage imbalance [24].

The grid-tied solar PV systems inject the active power with a unity power factor and thereby occasionally disturbing the balance of active and reactive power [25]. The increased usage of electric vehicles (EV) with this system can lead also to an increase in voltage imbalance [26]. Peak power curtailment (PPC) is used as a solution, but it has economic implications and it only helps with solving overloading in lines [5]. Several studies have proposed voltage control mechanisms based on increased reactive power demand from the substation and the high-frequency switching inverter can reduce voltage fluctuations but cause increased harmonics [27,28]. The on-load tap changer (OLTC) and feeder control voltage capacitor banks have also been proposed for the overvoltage problem [29,30]. Moreover, a demand response (DR) method [31], battery energy storage system (BESS) [32,33], and local flexibility markets [34,35] have been proposed for increased HC.

An active management solution has been proposed to operate PVs in feasible regions to avoid congestion [36]. Instead of solutions like line reinforcement, network reconfigurations and RPC, a congestion management market has been proposed which is based on the zonal pricing and flexibility market [37]. The intraday market prices and grid balancing services could also provide help in resolving congestion in the networks [38]. Several optimization algorithms are also proposed for the BESS scheduling problem including stochastic programming (SP) [39], model predictive control (MPC) [40] and robust optimization (RO) [41]. These algorithms do not cover all non-linearities and patterns in the datasets; therefore, the techniques lack in providing optimal solution for this optimal scheduling problem [42]. Machine Learning (ML) algorithms are presented in the literature to increase the HC of the power distribution networks. In [43], a solution of community energy storage based on reinforcement learning (RL) is proposed. This method requires additional sensors installed on the lines. Another RL-Monte Carlo-based method is also proposed for BESS management to mitigate overvoltage problems [44]. In this work, distributed control of BESS is used in other regions to provide voltage control. A battery size optimization technique

Table I
Comparative Analysis of this studies with existing literature.

Study	Country	Year	Congestion Control Mechanism					Applications			Economic Analysis
			PPC	RPC	OLTC	BESS	ML BESS	Increased HC	Voltage control	Line overloading control	
[47]	Netherlands	2023	×	√	√	√	×	×	√	√	×
[48]	Iran	2024	×	×	×	√	×	×	√	×	×
[49]	Columbia	2022	×	×	×	√	√	×	√	×	×
[5]	Estonia	2022	√	√	×	×	×	√	√	√	×
[50]	Greece	2022	×	×	×	√	×	×	√	√	×
[27]	USA	2017	×	×	√	×	×	×	√	×	×
[29]	Italy	2020	×	√	×	×	×	×	√	×	×
[51]	Sri Lanka	2022	√	√	√	√	×	√	√	×	×
[52]	Australia	2020	×	×	×	√	×	×	√	×	×
[53]	USA	2021	×	√	×	√	×	√	√	√	×
[30]	Netherlands	2015	√	×	×	×	×	√	√	×	×
[31]	Australia	2018	×	×	√	×	×	×	√	×	×
[54]	Portugal	2024	√	√	×	×	×	√	×	×	×
[32]	Sweden	2019	×	×	×	√	×	√	√	×	×
[33]	Australia	2020	×	×	×	√	×	√	√	×	×
[21]	China	2023	×	×	×	×	×	√	√	×	×
[55]	Turkey	2024	×	×	√	×	×	√	√	×	×
This paper	Estonia	2024	√	√	×	√	√	√	√	√	√

is presented in [45] to limit the PV curtailment requirements for increasing PV-HC in the distribution networks and lowering congestion. A deep learning based scheduling for PV-BESS is presented in [46]. It utilizes proximity policy optimization (PPO) to provide grid support. A comparative analysis of this study with previous studies described in the literature is presented in Table I.

All the above-described literature illustrates that excessive PV installations surpassing the HC limits can cause an increase in operating voltage and overloading in distribution lines implying severe difficulty for the grid to operate within the defined operation and safety limits. The proposed solutions, such as PV curtailment and reactive power control do not solve these problems completely. A comprehensive solution that solves all these problems without economic implications is required to be established. Therefore, in this paper, different techniques are compared and then a localized energy management solution (EMS) with residential BESS is proposed after comparing different BESS control strategies. An ML-based solution decides when to charge and discharge BESS to solve overloading and overvoltage problems by supporting the grid in safer operation and making sure that the users would not get a significant economic hit in the process by optimizing the energy usage costs. This proposed BESS control algorithm can be implemented in the EMS, energy router, or smart invertors. The main contributions and novelties of the paper are as follows:

- This paper presents an MLC strategy for dis−/charging the BESS of a household to increase PV hosting capacity and decrease congestion in the distribution network. Moreover, an LSTM algorithm for day-ahead PV power generation forecasting is incorporated as it dynamically optimizes the charging and discharging cycles of BESS to balance the load and mitigate overloading and voltage issues. Furthermore, the proposed MLC method is compared with BTC and ADC based BESS control techniques to identify the best strategy. The comparison provides a thorough evaluation of the feasibility and effectiveness of each method in increasing PV hosting capacity and minimizing congestion.
- This study is carried out on a real-world suburban LV distribution network, which includes multiple residential homes with rooftop-installed solar PV systems. The proposed technique is tested on four different rated PV installation created scenarios, the PV hosting capacity and the number of hours of line overloading and over-voltage are evaluated by the power flow analysis.
- The concept of the bus importance index is introducing the critical points in the network. This index helps in identifying the most vulnerable lines in the network which can allow the network control operator for a targeted intervention to manage congestion and enhance grid reliability.
- A comprehensive year-long techno-economic analysis of all four different PV installation scenarios and three BESS control strategies is conducted to ensure the robustness and applicability of the proposed solutions under various conditions. The findings highlight the economic viability of the proposed MLC solutions, demonstrating significant improvements in hosting capacity, and grid stability without imposing substantial costs on consumers.

The rest of the paper is structured as follows: The proposed congestion control strategies are presented in Section 2. Section 3 provides details about the LV network under consideration and the PV installation scenarios. The results and discussion are given in Section 4 together with the economic analysis. Section 5 concludes the paper with future directions.

2. Congestion Control Techniques & Battery Management

In this paper, three congestion control schemes with different battery management techniques termed as (a) Battery Trivial Control (BTC), (b) Analytics Driven Battery Control (ADC), and (c) Machine Learning based Battery Control (MLC) are presented to determine the feasible case/scenario to provide increased PV hosting capacity and minimal congestion in the LV distribution network. This proposed mechanism is a part of the energy management system (EMS) platform that holds solar PV, battery storage and a specific energy router (ER) management unit. This is a hybrid inverter but with additional functionalities as shown in Fig. 2. The ER will also have an optional DC bus for the DC load inside the house to be utilized in a more energy-efficient manner. It will also be equipped with a Wi-Fi data transmission system by which the users can monitor the energy utilization and generation in the house using a mobile/web application. The ML methods are the basis for proposed battery control mechanisms that would be implemented in this ER. The ER continuously monitors the PV energy generation, electrical load and status of the battery and is equipped with a multitarget decision algorithm that decides when to charge/discharge the battery. Having the control capabilities beyond the in-house benefits will provide advantages, for example, considering local energy communities, small-area energy hubs, etc., where resources are dearest. Also, for the utilities providing the congestion relief service is seen as well-regarded future benefits. Further details of this proposed EMS can be found in [56].

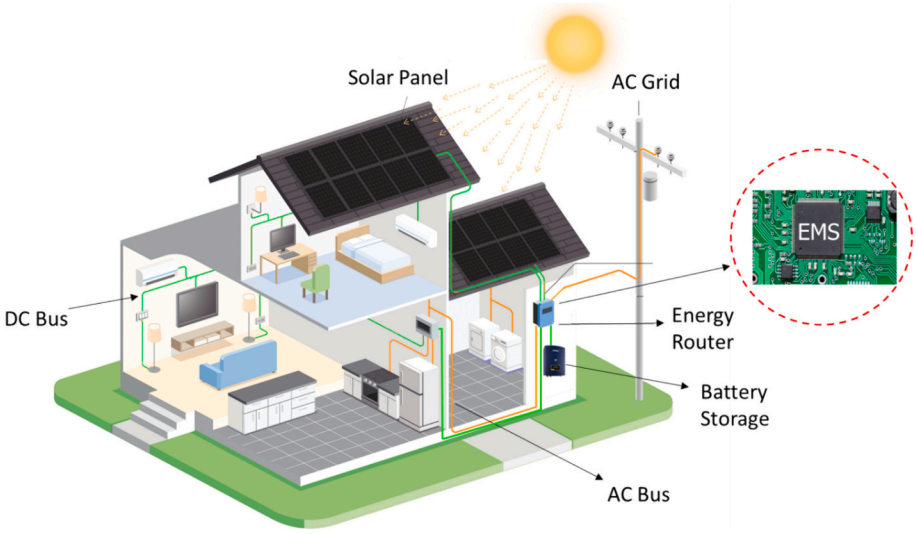


Fig. 2. The structure of the proposed energy management system.

The priority is to provide local control in the distribution network using BESS control, and PV power curtailment, and avoid installing extra devices, such as reactive power control and OLTCs. The reactive power control-based smart inverters solve the overvoltage problem but do not eliminate line overloading [5]. All three proposed congestion control schemes are discussed separately in this section. However, first, the bus importance index is computed to estimate the impact of increased power flow on the electrical lines and its impact on the bus-bars/nodes in the LV distribution network. This index helps to observe and analyze the sections in the system that are more vulnerable to failure due to limited HC and, congested electrical lines due to increased power flow, and over-voltage on busbars due to installed large solar PV power production.

2.1. Bus importance index

Index calculation relies on the estimate of the power flow ratio on each electrical line in the entire LV network, which is computed in Eq. (1).

$$P_{ratio} = \frac{P_i}{P_{max}} \tag{1}$$

where the P_i is the actual real power flow on ith electrical line and P_{max} is the maximum power flow that the ith electrical line can sustain. When power flow on any electrical line increases due to the excess solar PV generation on a nearby busbar/node or an electrical line failure occurs, the AC power flow on some of the electrical lines will increase and cause voltage instability. The measurement of the power flow ratio is useful for developing the bus importance index, which is a useful tool in power flow analysis as it works on the principle of centrality [57]. This index helps us to identify the busbars that are most vulnerable in the event of excessive power flow on the adjacent electrical lines. The method uses an eigenvector concept to determine the centrality value for every busbar in the network and also assign a value β for all isolated electrical lines in the network using the eq. (2) [58]:

$$C_i = \alpha \sum_j A_{ij} \frac{C_j}{k_j^{out}} + \beta \tag{2}$$

where A_{ij} is the adjacency matrix [57], C_j is the jth bus centrality, α is the damping factor that has a value between 0 and 1, and k_j^{out} is the out-degree of the bus. In the adjacency matrix, the entry A_{ij} is decided using the AC power flow ratio computed for the electrical line that is connecting bus i with bus j, as computed in Eq. (1). The out-degree defines the number of other buses directly connected to the jth bus there is no other bus directly connected to the jth bus, then the centrality value of that bus will be undefined. Therefore, to avoid such a scenario, we set $k_j^{out} = 1$ for all such buses and the Eq. (2) can be further simplified as:

$$C = \alpha A D^{-1} C + \beta 1 \tag{3}$$

where D is a diagonal matrix with elements as $D_{jj} = max\left(k_j^{out}, 1\right)$ and $\beta 1$ is a vector with all entries are one (1,1,1, ….) [27].

2.2. Peak power curtailment

In this technique, the upper limit for power output from the photo-voltaic (PV) panels is capped at a certain value sometimes 90%, 70%, or 50% [5,13], to address network congestion, instead of completely dis-connecting the user. The peak power is typically generated during midday, especially in the summer when residential loads are at their lowest. In this scenario, rather than injecting the entire generated power into the grid, only a specific percentage is permitted. This approach helps mitigate the peak of PV energy, effectively alleviating network congestion, as illustrated in Fig. 3. However, it is important to note that curtailing PV power will have economic consequences for all customers, as the amount of energy sold to the grid will be restricted [12].

2.3. Reactive power control (RPC)

The implementation of the RPC mechanism is instrumental in preventing overvoltage issues within the network. Typically, grid-integrated PV systems predominantly feed active power into the grid and lack provision for reactive power. Nonetheless, as active power flow varies in opposite directions at different times, an influx of excessive active power and vulnerabilities in the line cross-sections may give rise to voltage fluctuations. The remedy involves employing inverters with enhanced capabilities to absorb or generate reactive power as needed. However, integrating RPC into solar inverters can lead to reduced efficiency and, at times, an increase in harmonic distortions within the network. The outcomes of RPC are depicted in Fig. 4 [5]. It is important to mention that the RPC method greatly helps in reducing voltage problems in the network, however, due to the increased power flow on the lines the overloading of the line could become even worse [5].

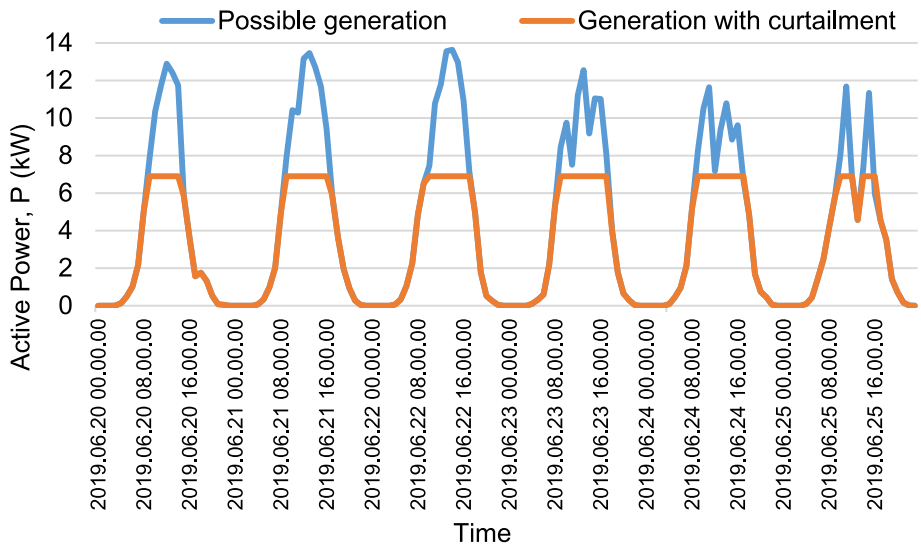


Fig. 3. The peak power curtailment.

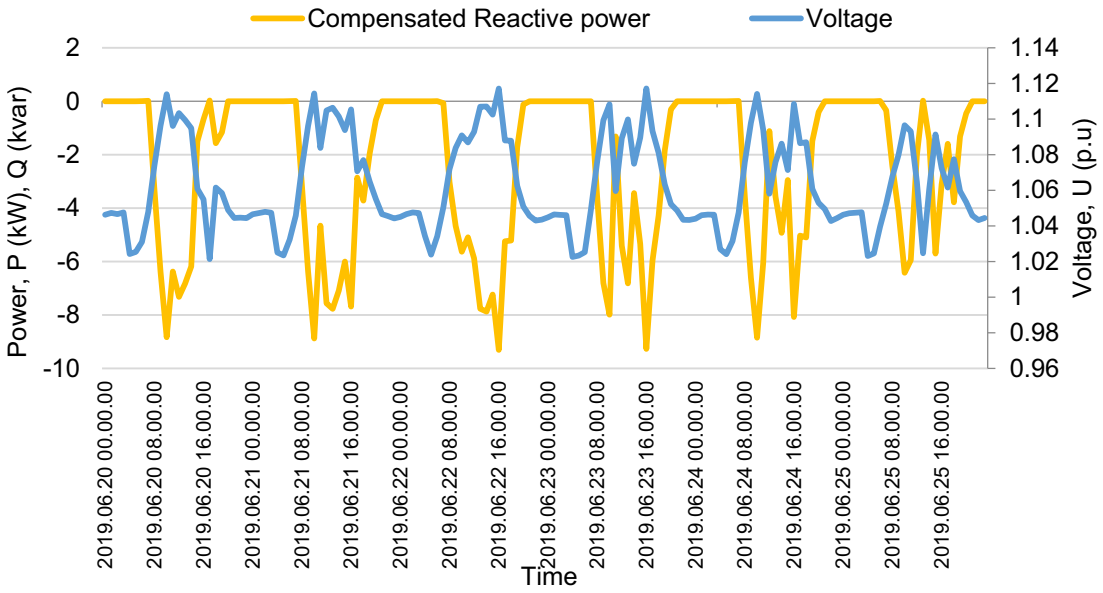


Fig. 4. Reactive power control.

2.4. Battery trivial control (BTC)

The BESS capacity for the installed solar PV system is computed using Eq. (4) as [59]:

$$E_{BC} = \frac{E_{Day}}{\eta^{*}DoD^{*}1000}.N_d \tag{4}$$

where the E_{BC} is the BESS capacity (in kWh), η is the BESS efficiency, N_d represents the number of days for which the BESS backup is required, the DoD is the Depth-of-Discharge for the BESS, and E_{Day} is the average daily energy usage in kWh. A valid assumption is taken that BESS capacity is sufficient to supply one full day and DoD is 70% [8]. The decision to charge and discharge the BESS is based on a heuristic algorithm proposed in [9,59]. The control algorithm is presented in Table II. Under the BTC scheme, the computed BESS capacities for the four rated PV system cases are given in the next section (Table III). The computed BESS capacities are expected to store the energy generated by the PV and to reduce the peak power infeed, to minimize the power delivery congestion in the LV network.

The performance analysis of the BESS control algorithm is conducted on the real-time PV generation and customer endpoint load data. In trivial, the simplest control sees if the PV power is higher than the load power, then BESS is charged; if vice versa, BESS energy is used to supply the customer's in-house load demand. The BESS operation is sustained with State-of-Charge (SoC) within specified limits between 20% and 90% of total energy capacity [5]. The SoC for the BESS is calculated on an hourly basis using Eq. (5):

$$SoC(t+1) = SoC(t) + \frac{\eta.P_z(n).K_p(n).t_s}{E_{BC}} \tag{5}$$

where t represents the index of an hour under consideration, t_s is the sampling interval in hours, K_p indicates the online and offline status (typically 0 or 1), and P_z is the BESS power in kW. Moreover, the algorithm also incorporates the market electricity price for charging the BESS and selling energy from the BESS to the grid. The prices for charging and discharging BESS are variable and depend on the season. This algorithm uses linear programming (LP) and convex optimization to minimize the overall cost of energy utilization. The objective function

Table II
The Proposed Algorithm.

Algorithm: Proposed Battery Charging & Discharging
Input: Load data, PV data, Electricity Price

1. Calculate battery E_{BC}, E_z
2. $SoCmax = 0.9$ and $SoCmin = 0.2$
3. **While ($n \leq 8760$)** %n is the number of hour
4. **If** $PV > Load$ and $0.2 > SoC\ (n) > SoCmax$ then charge the battery (But not above $SoCmax$)
5. Calculate $SoC\ (n+1)$ and $PBat\ (n)$ % no. of charging hours with PV
6. **else if** $Load > PV$ and $SoCmin > SoC\ (n) > SoCmax$ then discharge battery (But not below $SoCmin$)
7. Calculate $SoC\ (n+1)$ and $PBat\ (n)$ % no. of discharging hours for internal use
8. **else if** Electricity Price < threshold value and $SoC(n) < 0.5$ then Charge battery from the grid (But not above $SoCmax$)
9. Calculate new $SoC\ (n+1)$ and $PBat$ % no. of charging hours with Grid
10. **else if** Electricity Price > threshold value and $SoC(n) > 0.4$ then discharge battery to the grid (But not below $SoCmin$)
11. Calculate $SoC\ (n+1)$ and $PBat\ (n)$ % no. of discharging hours to the grid
12. **else**
13. $SoC\ (n+1) = SoC\ (n)$
14. $PBat\ (n) = 0$
15. **end if**
16. $n = n + 1$
17. **end while**

Table III
Load and PV profiles.

Users	Peak load (kW)	PV installed capacity (kW)				BESS capacity (kWh)				Line parameters		
		Case 1	Case 2	Case 3	Case 4	Case 1	Case 2	Case 3	Case 4	Lines	Nom. current (A)	Length (m)
L1	3.0	13.8	5.0	5.0	4.4	42	16	16	14	VKS F03_0	605	60
L2	4.8	11	6.8	6.8	6.0	34	22	22	18	VKS F03_1	115	342
L3	3.7	11	5.7	5.7	4.1	34	18	18	14	VKS F03_2	140	44
L4	0.7	4.2	2.7	2.7	0.7	14	10	10	4	VKS F03_3	90	23
L5	6.6	11	8.6	8.6	3.9	34	26	26	12	VKS F03_4	70	11
L6	5.8	13.8	7.8	7.8	4.4	42	24	24	14	VKS F03_a	140	45
L7	3.1	13.8	5.1	5.1	2.7	42	16	16	10	VKS F03_b	140	79
L8	3.4	13.8	5.4	5.4	4.1	42	18	18	14	VKS F03_c	140	38
L9	2.2	13.8	4.2	4.2	2	42	14	14	8	VKS F03_d	140	38
L10	2.3	13.8	4.3	4.3	2.1	42	14	14	8	VKS F03_e	140	44
L11	4.1	13.8	6.1	6.1	2.8	42	20	20	10	VKS F03_f	140	39
L12	1.3	13.8	3.3	3.3	1.6	42	10	10	6	VKS F03_g	90	102
L13	3.4	13.8	5.4	5.4	3.9	42	18	18	12	–	–	–
L14	4.0	13.8	6.0	6.0	1.3	42	16	16	4	–	–	–
L15	4.2	13.8	6.2	6.2	5.2	42	20	20	16	–	–	–
Total installed PV capacity	–	190	82	132	50	–	–	–	–	–	–	–

is defined as [59]:

$$Minimize\ f = \sum_{t=1}^{N} \left\{ C_{grid}^{pur,u}(t).\widehat{E}_{grid}^{pur}(t) + C_{grid}^{sell,u}(t).\widehat{E}_{grid}^{sell}(t) + C_{bat}^{ch,u}(t).\widehat{E}_{bat}^{ch}(t) + C_{bat}^{dis,u}(t).\widehat{E}_{bat}^{dis}(t) + C_{PV}^{u}(t).\widehat{E}_{PV}(t) \right\} \quad (6)$$

Subject to

$$E_{grid}^{pur}(t) - E_{grid}^{sell}(t) - E_{bat}^{ch}(t) + E_{bat}^{dis}(t) + E_{PV}(t) = E_L(t) \quad (7)$$

$$-E_{bat}^{ch}(t) + E_{bat}^{dis}(t) \leq E_L(t) \quad (8)$$

$$E_{grid}^{pur}(t) - E_{grid}^{sell}(t) \leq E_L(t) \quad (9)$$

$$-E_{bat}^{ch}(t) + E_{bat}^{dis}(t) \leq E_{bat}^{max}(t) \quad (10)$$

$$E_{bat}^{ch}(t) - E_{bat}^{dis}(t) \leq E_{bat}^{max}(t) \quad (11)$$

$$-E_{grid}^{pur}(t) \leq 0 \quad (12)$$

$$-E_{grid}^{sell}(t) \leq 0 \quad (13)$$

$$-E_{bat}^{ch}(t) \leq 0 \quad (14)$$

$$-E_{bat}^{dis}(t) \leq 0 \quad (15)$$

$$-E_{PV}(t) \leq 0 \quad (16)$$

$$E_{PV}(t) \leq E_{PV}^{max}(t) \quad (17)$$

where $E_{grid}^{pur}(t)$ and E_{grid}^{sell} are electrical energy purchased from and sold to the grid [kWh], respectively; the E_{bat}^{ch} and E_{bat}^{dis} are the electrical energy supplied to and extracted from the BESS [kWh], respectively; the E_{PV} is the electrical energy generated by the installed PV system [kWh]; N is the total number of hours in one year [8760]; $C_{grid}^{pur,u}$ is the per-unit

electrical energy purchasing cost from the grid [cents/ kWh]; $C_{grid}^{sell,u}$ is the per-unit electrical energy sell cost to the grid [cents/kWh]; $C_{bat}^{ch,u}$ is the per-unit BESS charging cost [cents/kWh]; $C_{bat}^{dis,u}$ is the per-unit BESS discharging cost [cents/ kWh]; $C_{PV}^{u}(t)$ is the per-unit cost incurred from the PV system [cents/kWh]. The hat symbol used with the electrical energies in the optimization cost function denotes the normalized values of the variables. All the constraints of the cost function are mentioned in Eq. (6) are defined in detail in our previous study [59].

With intense PV generation power available for multiple days, the trivial battery control has its limitations in reducing network congestion. That is, the BESS will be mostly fully charged during the morning hours and will not use the energy of the peak PV hours. Thus, the BESS with trivial control will not be able to decrease any PV infeed peaks.

2.5. Analytics driven battery control (ADC)

In this proposed energy management strategy, the BESS is controlled by the user, however, it has mild support for the utility functions. The main control aims to utilize the on-site PV-generated energy surplus, especially at the peak generation hours to charge the BESS with high power. The BESS in this scenario will be able to reliably shave off the high peaks of power injection into the network. This helps to combat congestion in the LV distribution network during peak hours of PV energy generation, for example, from midday to noon.

The control algorithm proposed earlier is used as a base and is further improved with the BESS start of charge time setting so that the BESS charging starts only during the peak energy generation hours. Furthermore, the BESS stored energy is sold to the grid in the evening hours when the electricity prices are comparatively higher, to the extent that is not expected to be used in-house during the nearest times. In this scenario, the BESS is not charged from the grid at any time interval. PV generation data and grid power flow analysis initially showed that peak overloading in electrical lines occurs in the middle of the day from 10 AM to 3 PM. In Nordic regions, overload is expected to occur mostly in the summertime when PV power generation is at its peak and the load is at its lowest. Therefore, the algorithm is tailored to charge BESS in the peak power hours and discharges the BESS to sell energy to the grid later in the evening during summer.

2.6. Machine learning based battery control (MLC)

In this method, a machine learning (ML) based forecasting algorithm is developed to provide a 24-h advance on PV generation levels. This forecasting will help to dynamically charge/discharge the BESS rather than on fixed times, as proposed in ADC. This method can provide a more optimal solution in terms of energy-cost reduction for customer congestion avoidance in the network. The BESS is charged, and the algorithm predicts the peak energy for the day, and it is discharged in the evening time when the electricity market prices are higher. In Fig. 5, the comparison of this BESS with MLC mechanism with previous BESS-based TBC is shown.

The long-short-term memory (LSTM) network algorithm is used here which is based on recurrent neural networks (RNN) which are categorized as deep learning (DL) algorithms [60,61]. This algorithm is most suitable for time series-based forecasting problems as it incorporates current and past data (stored in memory cells) for future forecasting [62]. This allows the memory units in LSTM to enable better handling of abstractions in long-time series data, as compared to traditional RNNs. A limitation of conventional RNNs is that they struggle to learn dependencies between current values and long-term data points because of the "vanishing gradient" issue. LSTMs overcome this issue by keeping, updating, or deleting cell data after each iteration. The architecture of an LSTM cell is depicted in Fig. 6. The LSTM architecture has three gates in it: input, forget, and output gates [63]. The gates in LSTM use a sigmoid function to get an answer of updating or forgetting the information and are represented by these Eqs. [64]:

$$i_t = \sigma(w_i[h_{t-1}, x_t] + b_i) \tag{18}$$

$$f_t = \sigma(w_f[h_{t-1}, x_t] + b_f) \tag{19}$$

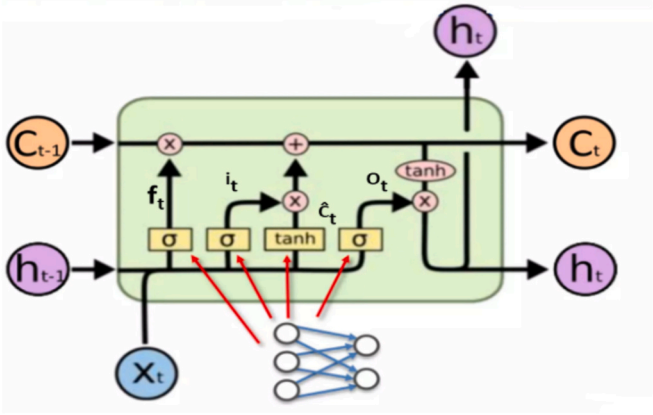


Fig. 6. The architecture of an LTSM cell.

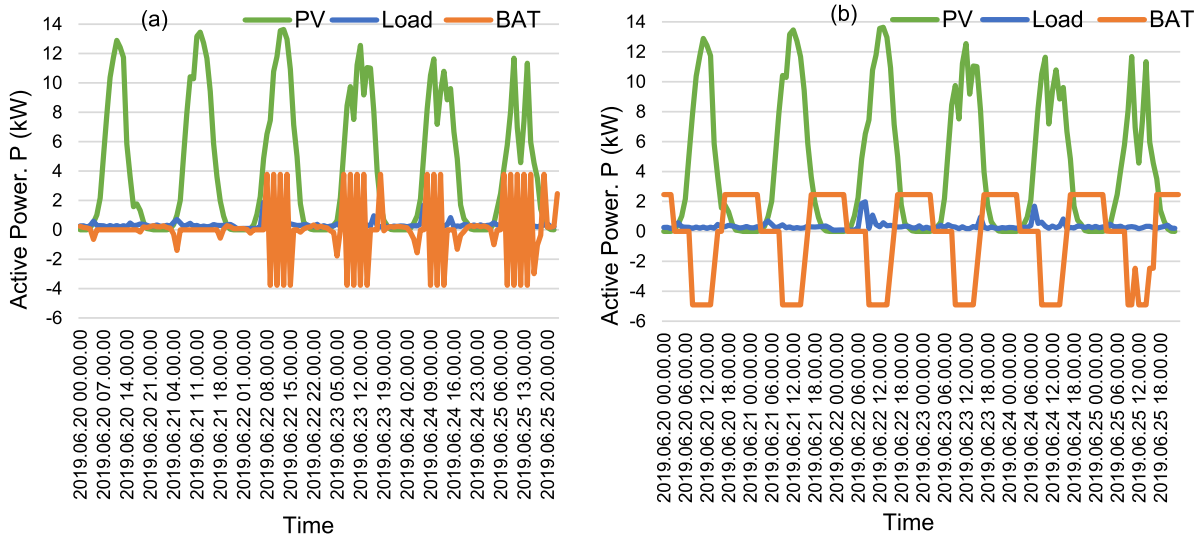


Fig. 5. (a) BESS with TBC (b) BESS with MLC

$$\hat{C}_t = tanh(w_c[h_{t-1}, x_t] + b_c) \quad (20)$$

$$C_t = f_c{}^*C_{t-1} + i_t{}^*\hat{C}_t \quad (21)$$

$$h_t = o_t{}^*tanh(C_t) \quad (22)$$

$$o_t = \sigma(w_o[h_{t-1}, x_t] + b_o) \quad (23)$$

Here, i_t is the input gate, o_t is the output gate and C_{t-1} is the input from the memory cell. The sigmoid function is represented by σ, w is the respective gates neurons, x_t is the input and h_t is the output values at the current timestamp t and b are the respective gate biases. The decision to store the current data in the memory cell is made by the sigmoid function based on h_{t-1} and x_t values by outputting either 1 or 0 in the cell state C_{t-1}. In the next step, a 'tanh layer' vector is created for new values $\hat{C}_t$ that could be added to the state. Then if stored earlier, the data from the memory cell is taken into consideration and the new cell state C_t is updated from C_{t-1}. The old state is multiplied by the forgetting state f_t (if the data was forgotten earlier) and added in $i_t{}^*\hat{C}_t$ to generate the new candidate value C_t. Finally, x_t and h_{t-1} are combined to decide what part of memory is going to the output module.

Historical datasets of PV energy generation and solar irradiance (x_t) for one year with a one-hour frequency have been used in the development of this algorithm. We used sequential layer architecture for our LSTM model that contains four LSTM layers with each layer containing 50 units (neurons) with a drop-out of 0.2 and one output layer with 1 unit. The Adam optimizer is used for the training of the LSTM network with mean square error as a loss function. The dataset is split into a 70:30 ratio for training and testing. In the training for all the inputs, a data structure is generated with 24 timesteps (hours) and one output.

3. Distribution network under study and rated PV systems

For the implementation and analysis of the proposed congestion control schemes, we consider an actual low-voltage DN line segment from a suburban Estonian grid. The part of the LV network taken for the study is a 0.4 kV feeder, which is supplied from the large distribution network of MV through a - 10/0.4 kV transformer in a substation. The actual measured data of the DN is gathered and then remodeled in DigSilent Power Factory 2022 including the existing 10 kV line and three detailed feeders of 0.4 kV each grid connected to the distribution substation. The load and PV production data are also yearlong recorded for the same LV network. These real-time recorded data sets make it possible to replicate actual power flow within the model and voltage fluctuations. The single-line diagram of the LV network under consideration is depicted in Fig. 7. There are fifteen users in this line segment and their load profiles are shown as a box plot in Fig. 8.

The nominal value of the switch breaker for the feeder F03–0.4 kV is 200 A. The detailed parameters of the lines are also given in Table III. The substation transformer rating is 100 kVA, and the nominal value of the switch breaker is 100 A. The load capacity of the cable AXMKA 3 × 50, used in the largest part of the constructed line, is 250 A. The sum of the main fuses of the customers on the line is 345 A. Despite the

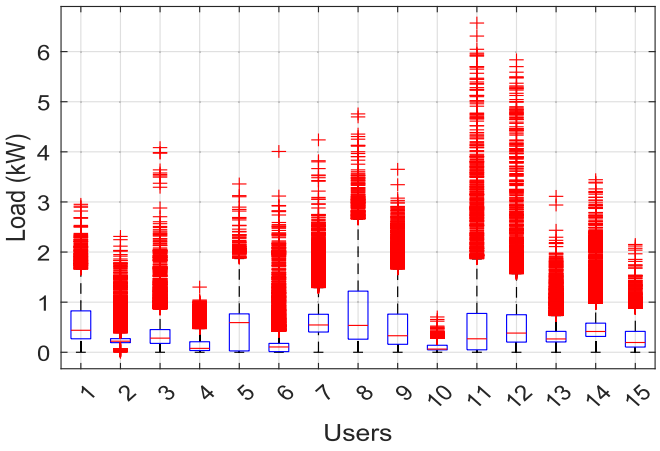


Fig. 8. The Box plot for all the loads.

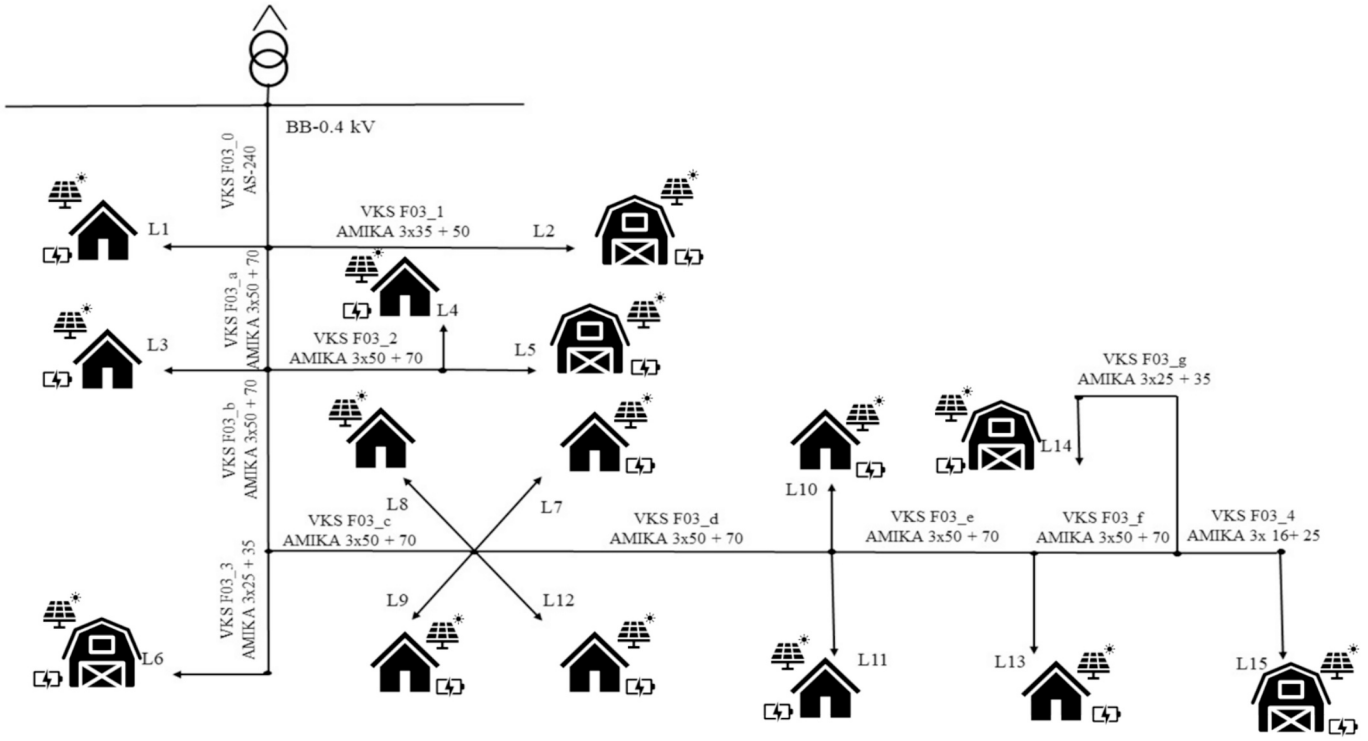


Fig. 7. The LV network segment under consideration.

relatively small power demand at the

customers' connection points, the nominal values of the main fuses vary from 10 A to 25 A with 3-phase connections installed. Due to this fact, there is a possibility that customers may install PV systems, rated according to the nominal current fuses. When the total capacity of PV systems was 238 kW, the whole LV network would be heavily overloaded.

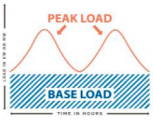
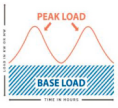
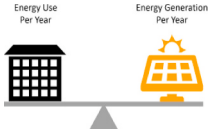
The electrical load profiles and PV power generated were measured in 2019. The solar panel values in some cases have.

been scaled to the value of the installed capacity of individual PV systems presented earlier in Fig. 1(b). The initial analysis shows that for such scenarios, the grid faces several problems, such as overloading of electrical lines and transformers and overvoltage problems in most of the nodes/buses, as shown in Fig. 6. From a detailed analysis of the testing results and the equipment data, it is evident that sections VKS_F03_a and VKS_F03_b (Fig. 6) have the largest electrical load in the network. The rated current of the corresponding overhead lines is 140 A and the maximum hosting capacity of these sections is 96 kW, which is near the transformer's nominal value of 100 kVA [5].

3.1. Case 1 (max. Profit)

The description of all the cases under consideration is given in Table IV. The highest-rated PV system installation scenario in this analysis is called *Installation Case 1 / 'Max profit'*, where a customer aims for the highest income profit from the endpoint in its capabilities. In *Case 1*, it is considered that the PV system's nominal power will be one step down from the nominal fuse rating. This means that if the connection point has a contract for 25 A fuses, the installed PV capacity will be 13.8 kW (that is, 20 A). Therefore, the total installed capacity of all 15 PV systems is 190 kW.

Table IV
PV Installation scenarios under evaluation.

Scenario	Name of scenario	PV Installation	Description
Case 1	Max. Profit	 PCC Fuse Rating	Installed PV according to the rated current of the fuse at point of common coupling (PCC).
Case 2	Peak load		PV rating according the individual user's peak load in the year.
Case 3	Additional PV	 	Individual user PV rating same as Case 2 but an addition 50 kW PV installed directly to the substation.
Case 4	Net zero energy		Installed PV rating according the energy requirement of the user for the whole year. (net zero energy building)

3.2. Case 2 (peak load)

The second scenario for the installation of the PV system is named *Installation Case 2 / "close to maximum power"*. *Case 2* is a more end-user normal power demand supporting case, being a less-stress case for the grid. In *Case 2*, the installed capacity of PV systems based on the endpoint maximum load power value [5].

3.3. Case 3 (peak load with additional PV)

A modification is made in *Case 2* and the third PV system installation case is named as: *Installation Case 3" Close to max power with added large PV*", the previous network PV installation case (*Case 2*) is used as a base, and we added a large PV power producer of 50 kW connected directly to the LV busbar at the substation. This case is introduced to investigate the LV network operation with a raised large PV total capacity in the LV feeder, as in this case leverage to the end furthest line end customer overvoltage problem is available.

3.4. Case 4 (net zero energy)

The last scenario investigated is named as: Installation Case *4 "Net Zero electric energy"* where the installed power selection for the PV system is based on annual electric energy consumption. The PV installation power is provided for annual total on-site PV energy generation to match the on-site electric energy utilization.

4. Results & discussions

This section describes the technical and economic impact of the proposed strategies. All the designed BESS control algorithms along with grid modeling and power flow analysis are carried out in DigSilent Power Factory for testing, and validation in real-time.

4.1. Centrality measure for the system

Fig. 9 shows the nodes of the LV distribution network earlier presented in Fig. 7. It is the box plot of the system's busbars that represent the expected centrality value of the buses for the whole year. Additionally, the 'Red Plus' signs show the variance in the centrality values for each bus. Buses with higher centralities are centrality-based bus importance indices for the busbars/ more prone to cause electrical line failures because these buses are associated with electrical lines that have high AC power, and the load power and solar photovoltaic generation on these busbars are higher compared to other buses in the system. In Fig. 9, we observe that the busbars 'F03_12' and 'F03_20' are the most critical buses in the LV distribution network under study. Therefore, to curtail HC, the solar PV generation on these busbars requires more attention.

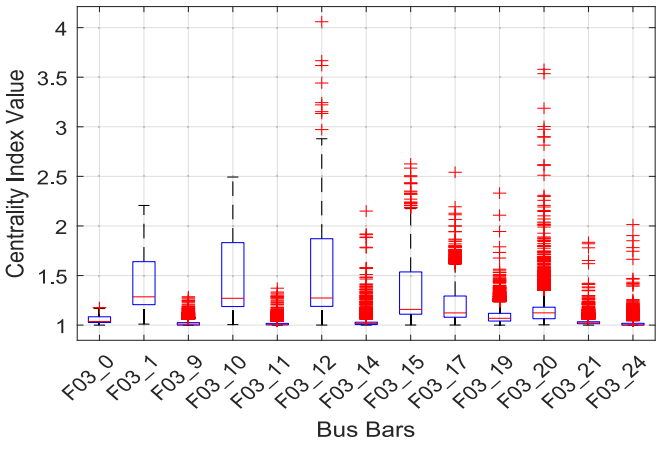


Fig. 9. The busbar importance indices.

4.2. Technical impact

The performance evaluation of three different techniques is implemented with BESS along with the RPC, and PPC is carried out here. For the overloading issue, grid reinforcement is usually required and for overvoltage problems, voltage control through smart inverters having some kind of RPC is one of the most common approaches. PV output curtailment and BESS installation are also common practices to reduce abnormal regimes. The grid reinforcement for the installed 189 kW would require a necessary replacement of the transformer for at least 200 kVA rating and electrical lines up-gradation from AMKA $3 \times 50 + 70$ to AMKA $3 \times 120 + 95$. The implementation of voltage regulation through smart inverters gives an improvement in overvoltage profiles. However, the loading of electrical lines and transformers is significantly increased as demonstrated in [5]. In Table V, the comparison of yearly PV utilized energy using various techniques is presented. This shows that curtailment reduces the PV HC significantly and it is not a good solution.

The RPC with smart inverters on the other hand resolves overvoltage issues, however, it worsens the line overloading. This can be explained by the increase in reactive power flow, which is used for voltage regulation. If no grid reinforcement is planned and the limitation of PV systems applies uniformly, usually each customer must reduce the output of the PV system at least twice and must take an economic hit. Therefore, the scope of this work is limited to reducing congestion with the installation of BESS and its efficient control. The duration of abnormalities is still the same in both the overload and overvoltage scenarios. The implementation of mild control (BESS with ADC) significantly decreases the overloading in the distribution lines to 15% and 25% for the transformer. In addition, there is a 25% improvement in the overvoltage problem computed and examined using Eq. (3). However, such kind of control is still not enough to make grid operation stable. The comparison of BESS control strategies for all cases is given in Table VI. In Table VI, the centrality measure concept is presented in Eq. (3) and shown in Fig. 8 is used to compute the electrical lines overloading and voltage overloading on busbars in the network for the whole year.

The comparative analysis of the four different congestion control techniques for Case 1(Maximum profit) is presented in Fig. 10. For Case 1 the installation of BESS with MLC gives the best results, it reduces the overvoltage problem significantly as the average value is lower and the peak is around 70% lower. The line overloading is almost eliminated as only the transformer and one-line segment are overloaded but only for a few hours throughout the year. Case 1 is exceptional in terms of PV installations; therefore, it requires a combination of BESS, RPC, and some curtailment as well to bring all these values into the normal operation range.

For Case 2, PV is installed close to peak load power. The results show that installation of BESS can significantly reduce voltage fluctuations and with the use of ADC and MLC strategies, it solves the hosting capacity problem both in terms of overloading and overvoltage.

In Case 3, the installation of BESS does not help solve the problems and increases the loading of the elements. However, more energy was sold to the grid rather than bought from the grid in the whole year. The overvoltage hours were reduced to four times as compared to the scenario without batteries. However, the ADC strategy again solves all the problems.

For net zero energy (Case 4), the installation of BESS even without control can fix the voltage issue, but to reduce the load on the transformer the implementation of a mild control strategy is necessary.

Table V
Yearly PV Utilization Using Different Techniques for Case 1.

Utilization scheme	E gen. (kWh)	Utilization
PV	224,106	100%
PV curtailment (70%)	156,874	70%
PV with BESS TBC	224,106	100%
PV with BESS MLC	224,106	100%
PV + RPC with BESS MLC	224,106	100%
PV curtailment (90%) with RPC and BESS MLC	214,099	96%

Table VI
Comparison of different BESS Control Strategies for all the.

		Without BESS			TBC			DAC			MLC		
		No. of hours	**Peak value**	**No. of Elements**	**No. of hours**	**Peak value**	**No. of Elements**	**No. of hours**	**Peak value**	**No. of Elements**	**No. of hours**	**Peak value**	**No. of Elements**
Case 1	Transformer overload (%)	698	200	1	689	159	1	170	134	1	**56**	**130**	**1**
	Lines overload (%)	558	176	4	556	159	3	84	119	1	**6**	**114**	**1**
	Nodes, U > 1.1 p.u.	1187	1.24	11	1154	1.2	11	880	1.18	10	**834**	**1.17**	**10**
Case 2	Transformer overload (%)	0	93	0	0	69	0	0	58	0	**0**	**57**	**0**
	Lines overload (%)	0	82	0	0	62	0	0	52	0	**0**	**50**	**0**
	Nodes, U > 1.1 p.u.	304	1.12	8	21	1.12	6	0	1.05	0	**6**	**1.04**	**0**
Case 3	Transformer overload (%)	254	141	1	211	117	1	141	111	0	**136**	**109**	**0**
	Lines overload (%)	0	81	0	0	62	0	0	52	0	**0**	**49**	**0**
	Nodes, U > 1.1 p.u.	434	1.13	8	117	1.12	8	0	1.09	0	**0**	**1.08**	**0**
Case 4	Transformer overload (%)	373	134	1	244	126	1	0	56	0	**0**	**56**	**0**
	Lines overload (%)	0	46	0	0	42	0	0	28	0	**0**	**27**	**0**
	Nodes, U > 1.1 p.u.	704	1.15	9	0	1.09	0	0	1.07	0	**0**	**1.05**	**0**

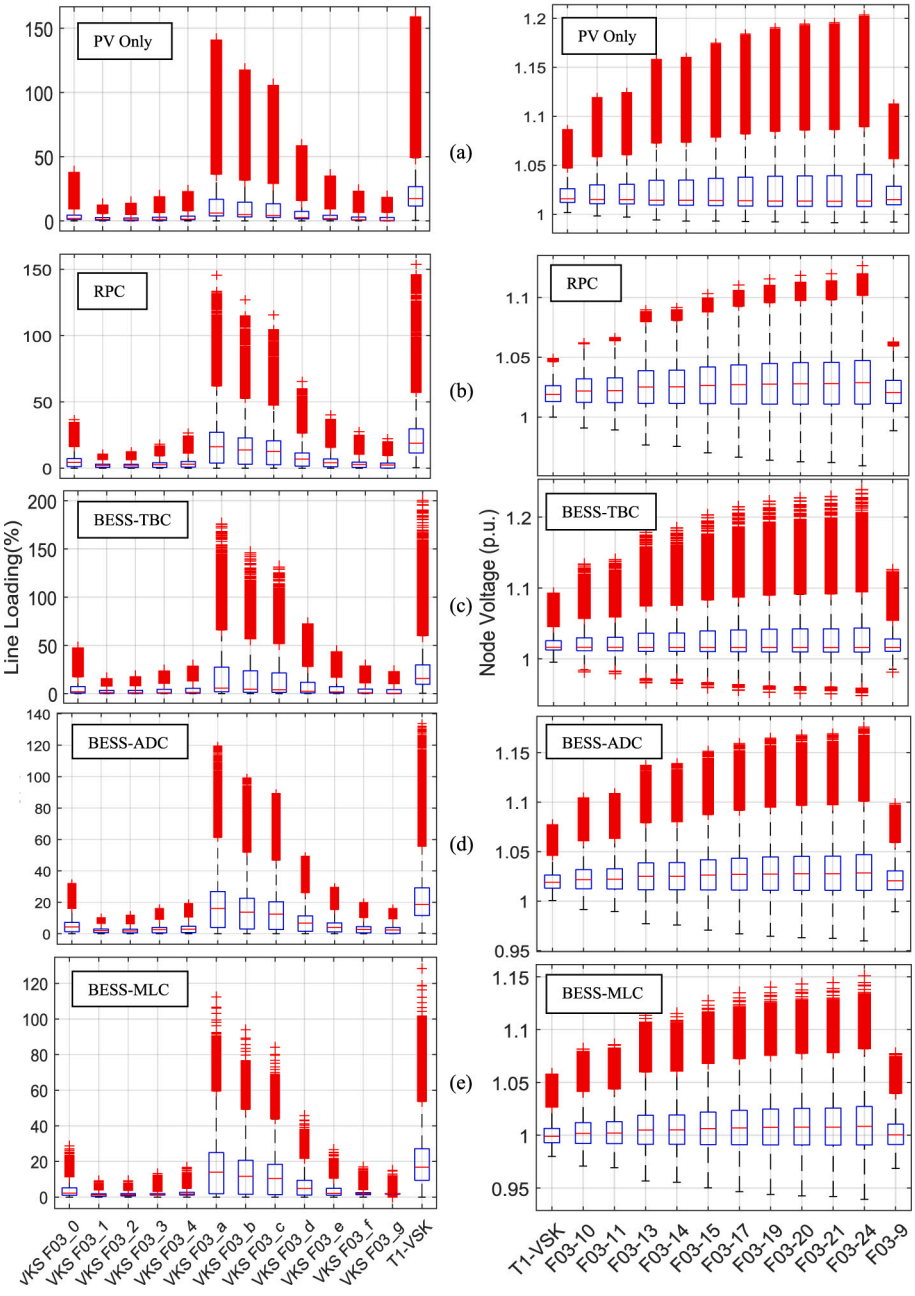


Fig. 10. Comparison of line loading and line voltage by employing different congestion techniques (a) with PV only (b) Reactive power control (RPC) (c) BESS with TBC (d) BESS with ADC (e) BESS with MLC.

4.3. Economic analysis

The economic analysis using both BESS control strategies and with BESS has also been carried out for all cases. The results of the economic analysis for all fifteen customers are presented in Table VII. The results indicate that PV installations significantly reduce the cost the energy for all the customers and the users can even earn money by selling energy to the utility company. The negative cost values in the table show that more energy was sold to the grid rather than bought from the grid in the entire year. Fig. 11 shows the market electricity prices in Estonia in 2021 and 2022. The electricity price in 2021 was on average around 0.03 € / kWh but in 2022 it was around 0.2 € / kWh [65]. The price increase is due to the geopolitical situation in the EU. There is also one hour where the price hit the ceiling of 4 € / kWh due to the unavailability of some power plants [65].

In all four cases of PV installation, all fifteen users would be able to make financial gains. In Case 1, as the installed PV-rated power is the highest here, the users will earn more money. Even with the PV installation only, all the users will have saving as surplus generated energy will be sold to the grid. The installation of BESS can further increase the revenue from the sale of this energy. In the MLC technique, the financial gains are the highest as compared to the other two control techniques. The saving varies between 8% to 14% for all the users. However, in the ADC control technique, in which there is a mild BESS control for congestion reduction, the gains are 6% to 8% more as compared to the TBC control method. In Case 2 and Case 3, the economic numbers for all the customers again show a similar pattern. The increase in financial gain varies from 11% to 22% for all the customers employing the MLC control strategy as compared to the TBC method. These numbers vary between 4% to 8% when compared with the ADC techniques. For Case 4,

Table VII
Economic Analysis For All The Case with 2021 prices.

Case 1					Case 2			
Users	1,3,5,9,11, 12,13,14, 15	2,4,6	10	7,8	1,3,5,9,11, 12,13,14, 15	2,4,6	10	7,8
OC	115–181	55–87	29	207–246	115–181	55–87	29	207–246
CPV	322–505	530–557	159	257–420	64–223	98–168	87	91–93
TBC	347–544	571–604	171	279–452	72–248	94–184	100	279–452
ADC	374–578	605–638	182	307–487	86–272	102–198	108	307–486
MLC	398–598	615–652	192	207–246	93–280	107–202	112	315–498
Case 3					**Case 4**			
OC	115–181	55–87	29	207–246	115–181	55–87	29	207–246
CPV	64–223	98–168	87	91–93	9–35	7–13	4	41–42
TBC	72–248	94–184	100	279–452	21–46	11–22	8	54–56
ADC	86–272	102–198	108	307–486	49–61	39–44	38	63–66
MLC	93–280	107–202	112	315–498	50–69	45–54	42	66–69

Abbreviations: Original cost (OC), Cost with PV only (CPV), Trivial BESS control (TBC), Analytics Driven Battery Control (ADC), Machine learning-based control (MLC). All costs are in Euros. All values described here in the table are negative indicating profit, except for the OC.

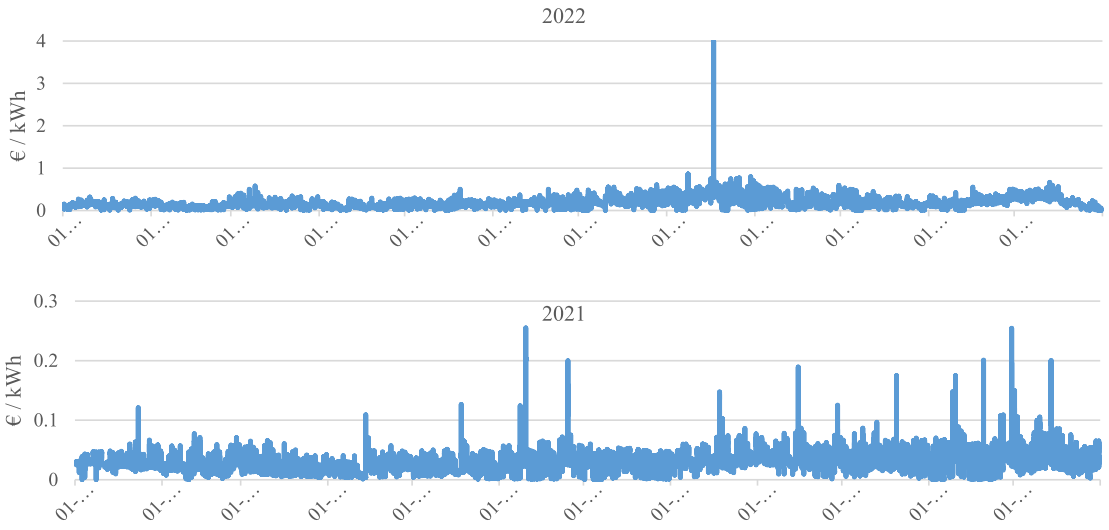


Fig. 11. The Estonian electricity market prices in 2021 and 2022.

the economic numbers verify that it was a nZEBs case as the value to financial gains for the users is not that high, only up to tens of euros. The economic difference between BESS control techniques MLC and TBC is also around ten euros. The economic analysis of a single household for all the different techniques is given in Fig. 12 based on the market prices of both 2021 and 2022. The MLC method gives better results in all scenarios. The prices of energy in 2022 are 3 to 4 times higher than in the year 2021 due to geopolitical reasons. Therefore, it shows more profit in 2022 if the methods described here are utilized.

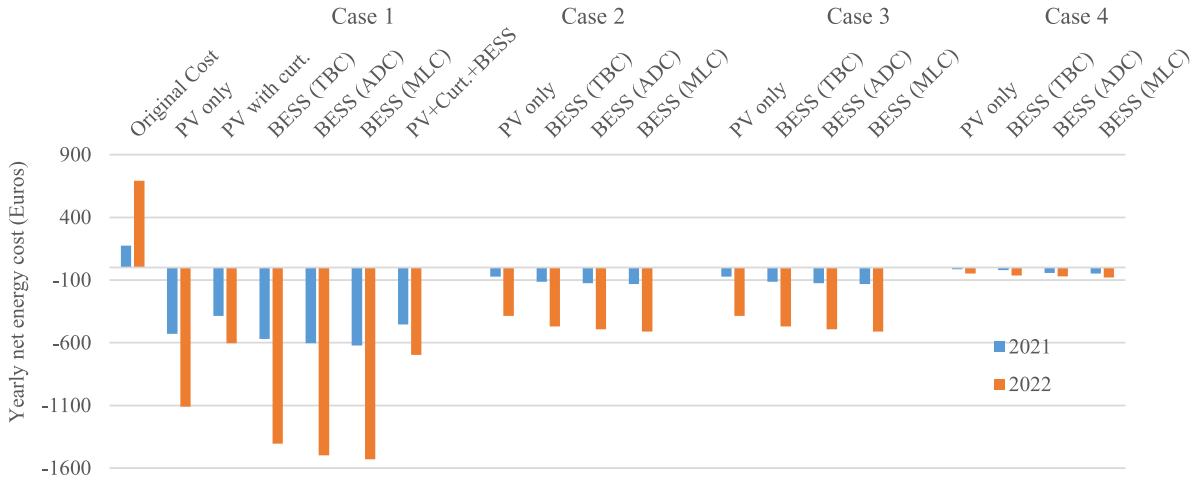


Fig. 12. Comparison of energy utilization cost of different techniques for 2021 and 2022.

5. Conclusion

This study examined various strategies to enhance the hosting capacity of the grid in response to the high penetration of the PV systems in the LV networks. While increased renewable energy penetration is desirable, it poses significant challenges for network operators, including potential bottlenecks leading to overloading and voltage fluctuations. High penetration of PV systems can cause overloading of distribution transformers and feeders, leading to voltage fluctuations and instability in the grid. The conventional solution may require grid reinforcements or additional devices for reactive power control.

This study evaluates four different PV installation scenarios on a real-life LV distribution network, providing practical insights into the challenges and solutions for different levels of PV penetration. The integration of a bus importance index effectively identifies critical points in the network, allowing for targeted congestion management and proactive infrastructure planning. By comparing BESS with BTC, ADC and MLC, the aim is to identify the most effective and economical control strategy for increasing PV hosting capacity and reducing congestion. The study conducts a detailed, year-long analysis of residential loads and PV generation, ensuring that the proposed solutions are robust and effective under varying conditions. The findings indicate that integrating PV systems with MLC-controlled BESS infrastructure can effectively increase HC and reduce energy service costs for residential users employing energy management systems EMS. This approach helps avoid costly network upgrades. The installation of BESS, combined with strategic control measures, mitigates the adverse effects of bidirectional power flows and congestion, thereby enhancing the grid's PV hosting capacity.

In scenarios of lower PV penetration (Cases 2, 3 and 4), installing BESS without advanced congestion control may suffice. However, at peak PV penetration levels, significant voltage fluctuations and overloading occur, severely limiting PV HC. While curtailing PV output can alleviate these issues, it results in a substantial 70% reduction in HC. Voltage regulation through smart inverters with RPC improves the voltage profile but increases equipment overloading due to higher reactive power flows. Additionally, BESS with trivial control does not adequately normalize grid operations under high PV penetration. The MLC-based BESS control mechanism reduces the overvoltage in the network by up to 30% as compared to BESS with TBC in Case 1. In these extremely large PV installations, combining BESS with RPC and minimal PV curtailment may still be necessary.

The techno-economic analysis underscores the economic viability of the proposed solutions, demonstrating significant improvements in hosting capacity and grid stability without imposing substantial costs on consumers. The.

By addressing these key challenges through innovative control strategies and comprehensive real-world analysis, the paper provides a practical and scalable solution for maximizing PV hosting capacity in distribution networks. The economic analysis also shows that the MLB-based method gives better savings as compared to all the other control schemes.

Future research should focus on implementing and evaluating these proposed schemes in a small nearly zero-energy district. Further, integrating electric vehicles into the scenario can provide additional insights into the effectiveness of these strategies.

CRediT authorship contribution statement

Noman Shabbir: Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Lauri Kütt:** Writing – original draft, Validation, Supervision, Methodology, Investigation, Conceptualization. **Victor Astapov:** Writing – original draft, Software, Resources, Investigation, Formal analysis, Data curation. **Kamran Daniel:** Visualization, Software, Resources, Data curation. **Muhammad Jawad:** Visualization, Validation, Software, Resources, Formal analysis. **Oleksandr Husev:** Writing – review & editing, Validation, Resources, Project administration, Funding acquisition. **Argo Rosin:** Writing – review & editing, Validation, Software, Formal analysis. **João Martins:** Writing – review & editing, Validation, Supervision, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work is supported by Estonian Research Council grant PRG675, the EU Commission through H2020 project FinEST Twins grant No. 856602 and by the project "Increasing the knowledge intensity of Ida-Viru entrepreneurship" co-funded by the European Union (2021-2027.6.01.23-0034).

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